

LIPIDS ARE ENDOGENOUS MODULATORS OF CLASS C GPCRS

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ABSTRACT

G Protein-Coupled Receptors (GPCRs) are modulated by membrane lipids, which help stabilize conformations, shape ligand pockets, and influence downstream signaling via G proteins and β -arrestins. However, how lipids modulate class C GPCRs, a sub-family of 22 obligate dimers with extensive membrane interactions, is not known. Yet, several observations suggest lipids may be key modulators of this family. First, recent cryo-EM structures revealed lipid densities at receptors' transmembrane domain (TMD) dimer interfaces, whose presence appear to be conformation-dependent^{1,2}. And second, class C receptors often localize in specific membrane nano-domains³, and their function is disrupted in lipid-altered cellular contexts, as seen in diseases like Niemann-Pick. However, the role lipids play in shaping class C GPCR function and the molecular mechanisms by which these modulations occur are still completely unknown. Here, I will present our latest work⁴ addressing these gaps-in-knowledge, mainly focusing on mGluR2 as a model of the family. Using a combination of cryoEM, nanobody- conformational sensors, nanodisc reconstitution and TIRF microscopy, we found that cholesterol is as negative allosteric modulator (NAM) of mGluR2, requiring multiple binding sites at its inactive TMD interface for an efficient functional modulation. We further show that this NAM effect is conserved across all mGlu, GABAB, and CaS receptors. Using all-atoms molecular dynamics combined with in-cellulo validation, we also identify two anionic lipid-binding sites that

independently act on mGluR2 conformation and recruitment of signaling partners. Finally, we investigate how their local lipid environment modulates how class C receptors interact with other membrane protein partners.

Together, our findings reveal that lipids are endogenous allosteric modulators of class C GPCRs⁴, regulating receptor conformational states at their TMD dimer interfaces and modulating their interactions with other proteins.

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